Spatial Variation and Influencing Factors of On-Street Parking Occupancy in Downtown Seattle, USA

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Preparation

- Initial the Datasset
- Number of blocks: 1470
- Number of Metered Meters: 18 196 (PTA), Spot 340
- Cleaning Metered Meters: 18 196 (with Unique Identifier), Pattern 23 515
- Combine the metered to block available with the Seattle Parking Occupancy
- Combine the metered to block available to GIS Dataset and select the needed
- Table

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1.0

Introduction

In high-density city centers, on-street parking provides convenient access to nearby facilities and transport options for residents without private spaces, but both over-occupancy and under-occupancy can cause congestion.

Existing studies show that parking occupancy varies by time, location, and land use. Commercial areas typically experience high occupancy during daytime hours, residential areas show different patterns between overnight and weekday use (Kingham et al., 2015), and that different parking types display spatial differentiation that can be explained by factors such as land use and street networks (Ma and Ma, 2019; Zhang et al., 2021).

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Research questions

This report takes four key real-world parking issues in downtown Seattle (**Bellevue, First Hill, Commercial Core and Pioneer Square**) as case studies to analyze street-level parking occupancy patterns, identifying spatial and time patterns and influencing factors.

RQ 1: What patterns of the block-scale parking occupancy in street-level parking across different streets in Seattle city center?

RQ 2: Are these different parking patterns spatially clustered?

RQ 3: What street-level and surrounding built environment factors can explain the variations in on-street parking occupancy rates in Seattle's city center?

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Data

Variable	Type	Description	Notes
Occupied spaces	Numeric	The number of parking spaces occupied in a street during a given period	Used for calculating occupancy rate
Paid parking	Numeric	The number of parking spaces available for paid parking in a given period	Used for calculating occupancy rate and as an explanatory variable (GIS)
Occupancy rate	Numeric	Proportion of total parking occupancy	Used to derive peak, low, (averaged), and (standard) occupancy
Block	Categorical	Distinguishes between the blocks	
Hourly rate	Numeric	Hourly parking fee in a given period	Used as an explanatory variable
Occupancy rate	Numeric	Street-level parking occupancy rate, calculated as occupied spaces divided by total parking spaces	Cross-scale analysis (block to GIS and PSA)
Business density	Numeric	Unique business per square kilometer	Used for examining accessibility to local amenities
Transit stop density	Numeric	Transit stops per square kilometer	Used as an explanatory variable
Distance to CBD	Numeric	Distance from parking block to the CBD	Used in correlation analysis
Distance	Text	Street name	Used to relate and analysis
Neighborhood	Categorical	Neighborhood name of the block/face	Study area: Bellevue, Commercial Core, Pioneer Square, First Hill
PlaceNameLocations	Categorical	Parking lot/ area within the parking meters	
Peak, low, (averaged), (standard) occupancy	Numeric	Derived from block-level occupancy rate	Used for spatial clustering (street segments)
Occupancy difference	Numeric	Difference between peak and low occupancy	Used as the dependent variable for OLS regression (clustering in PSA)
Street-level occupancy	Numeric	Street-level occupancy, aggregated from block to street segment	Dependent variable in RQ1
Street-level occupancy	Numeric	Street parking occupancy rate of each street segment across all time groups	Dependent variable in RQ2
Street-level occupancy	Numeric	Street parking occupancy rate of each street segment across all time groups	Dependent variable in RQ3
Commercial land use	Numeric	Share of land within the street buffer classified as commercial or retail use	Explanatory variable in RQ3
Residential land use	Numeric	Proportion of land within the street buffer occupied by residential	Explanatory variable in RQ3
Amenities	Categorical	Amenities classification of the street segment	Categorical explanatory variable in RQ3
Street length	Numeric	Length of each street segment in meters	Explanatory variable in RQ3
Street width	Numeric	Mean width of each street segment	Explanatory variable in RQ3
Transport parking share	Numeric	Share of land within the street buffer classified as transport or parking related use	Explanatory variable in RQ3
Intersection density	Numeric	Intersections per square kilometer within the street buffer	Explanatory variable in RQ3
Residential density	Numeric	Share of land within the street buffer classified as residential use	Explanatory variable in RQ3

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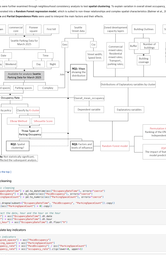
Methodology

As the parking occupancy data contain no coordinate-level data, **street-level block-facing** was used to identify patterns. Peak, (low), (averaged), (standard) occupancy were calculated and aggregated to block-level and street segment-level. Spatial clustering was then applied to the block-level data to identify spatial patterns, and OLS regression was used to examine the influencing factors at the block level (n = 1470) and street segment level (n = 2,304).

These types were further analysed through neighborhood consistency analysis to test **spatial clustering**. To explain variations in overall street occupancy, several variables were incorporated. Data from **Seattle Parking Occupancy (SPO)** and GIS is used to run linear multivariate and complex spatial clustering analysis (Kramer et al., 2021). **Remarkable importance** and **Partial Dependence Plots** were used to interpret the model factors and their effects.


Analysis
Dataset
Block-level data
Street-level data
Block-level occupancy
Street-level occupancy
Block-level occupancy rate
Street-level occupancy rate
Block-level occupancy difference
Block-level occupancy difference
Block-level occupancy difference

```
0 weekday_day_home-weekend_weekday_night-weekend_weekday_day_home-weekend_weekend_night-0.0
1 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
2 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
3 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
4 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
5 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
6 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
7 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
8 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
9 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
10 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
11 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
12 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
13 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
14 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
15 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
16 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
17 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
18 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
19 weekday_day_home-weekend_weekday_night-weekend_weekend_night-weekend_weekend_night-0.0
```

5.1.5. Figure 5: Distribution of occupancy rates for stream order stations in all four periods

Figure 5 provides a more detailed view of the median, dispersion and the range of occupancy rates for individual stream flow bar charts.

Cool among tested classified as intermittent stress at four time periods, occupancy rates on weekend nights remain significantly higher overall than during the other three periods, while rates during the other periods are similar. This suggests that, there is little variation, except observed for pairing at night and at weekends.

```
[nan, 0.0]  Distribution of Occupancy Rate for Stream Order Stations in All Four Periods
stream_order_cool_intermittent_stress_weekday_day_home_occupancy_rate_mean_weekend_weekday_night
stream_order_cool_intermittent_stress_weekday_night_weekend_weekend_night
stream_order_cool_intermittent_stress_weekend_weekday_night_weekend_weekend_night
stream_order_cool_intermittent_stress_weekend_weekend_night_weekend_weekend_night
count    20.000000  20.000000  20.000000  20.000000
mean      0.000000   0.000000   0.000000   0.000000
std       0.000000   0.000000   0.000000   0.000000
min       0.000000   0.000000   0.000000   0.000000
25%       0.000000   0.000000   0.000000   0.000000
50%       0.000000   0.000000   0.000000   0.000000
75%       0.000000   0.000000   0.000000   0.000000
max       0.000000   0.000000   0.000000   0.000000
[nan, 0.0]  Distribution of Occupancy Rate for Stream Order Stations in All Four Periods
stream_order_cool_intermittent_stress_weekday_day_home_occupancy_rate_mean_weekend_weekday_night
stream_order_cool_intermittent_stress_weekday_night_weekend_weekend_night
stream_order_cool_intermittent_stress_weekend_weekday_night_weekend_weekend_night
stream_order_cool_intermittent_stress_weekend_weekend_night_weekend_weekend_night
count    20.000000  20.000000  20.000000  20.000000
mean      0.000000   0.000000   0.000000   0.000000
std       0.000000   0.000000   0.000000   0.000000
min       0.000000   0.000000   0.000000   0.000000
25%       0.000000   0.000000   0.000000   0.000000
50%       0.000000   0.000000   0.000000   0.000000
75%       0.000000   0.000000   0.000000   0.000000
max       0.000000   0.000000   0.000000   0.000000
```

Figure 5: Distribution of Occupancy Rates for Stream Orders calculated in All Four Periods

Figure 5: Zoomed Scatter of Occupancy Rates for Stream Orders calculated in All Four Periods.

5.1.6 Chordal Graph for streams

A means examining how the pairwise relationship between stream types of stream performance prove to spatial analysis, in order to ascertain whether or more natural grouping structures emerge within the various:

- Stream order
- Intermittent stress, intermittent stress, non-stress & low-periodic
- Groundwaterbase: 0 stress
- Groundwater: stress-subsidence-allowable stress

```
# 5.1.6. Chordal Graph for Streams
# Visualizing pairwise relationships between top two per stream, with the occupancy rate for the current period
stream_order_cool_intermittent_stress_weekday_day_home_occupancy_rate_mean_weekend_weekday_night
stream_order_cool_intermittent_stress_weekday_night_weekend_weekend_night
stream_order_cool_intermittent_stress_weekend_weekday_night_weekend_weekend_night
stream_order_cool_intermittent_stress_weekend_weekend_night_weekend_weekend_night
# 5.1.6. Stream groundwaterbase
stream_order_groundwaterbase_stress_weekday_day_home_occupancy_rate_mean_weekend_weekday_night
stream_order_groundwaterbase_stress_weekday_night_weekend_weekend_night
stream_order_groundwaterbase_stress_weekend_weekday_night_weekend_weekend_night
stream_order_groundwaterbase_stress_weekend_weekend_night_weekend_weekend_night
# 5.1.6. Stream groundwater
stream_order_groundwater_stress_weekday_day_home_occupancy_rate_mean_weekend_weekday_night
stream_order_groundwater_stress_weekday_night_weekend_weekend_night
stream_order_groundwater_stress_weekend_weekday_night_weekend_weekend_night
stream_order_groundwater_stress_weekend_weekend_night_weekend_weekend_night
# 5.1.6. Stream 0 groundwater
stream_order_0_groundwater_stress_weekday_day_home_occupancy_rate_mean_weekend_weekday_night
stream_order_0_groundwater_stress_weekday_night_weekend_weekend_night
stream_order_0_groundwater_stress_weekend_weekday_night_weekend_weekend_night
stream_order_0_groundwater_stress_weekend_weekend_night_weekend_weekend_night
# 5.1.6. Stream 0 groundwaterbase
stream_order_0_groundwaterbase_stress_weekday_day_home_occupancy_rate_mean_weekend_weekday_night
stream_order_0_groundwaterbase_stress_weekday_night_weekend_weekend_night
stream_order_0_groundwaterbase_stress_weekend_weekday_night_weekend_weekend_night
stream_order_0_groundwaterbase_stress_weekend_weekend_night_weekend_weekend_night
# 5.1.6. Stream 0 subsidence
stream_order_0_subsidence_stress_weekday_day_home_occupancy_rate_mean_weekend_weekday_night
stream_order_0_subsidence_stress_weekday_night_weekend_weekend_night
stream_order_0_subsidence_stress_weekend_weekday_night_weekend_weekend_night
stream_order_0_subsidence_stress_weekend_weekend_night_weekend_weekend_night
# 5.1.6. Stream 0 allowable
stream_order_0_allowable_stress_weekday_day_home_occupancy_rate_mean_weekend_weekday_night
stream_order_0_allowable_stress_weekday_night_weekend_weekend_night
stream_order_0_allowable_stress_weekend_weekday_night_weekend_weekend_night
stream_order_0_allowable_stress_weekend_weekend_night_weekend_weekend_night
```

5.1.7 Figure 6: Chordal Graph for Streams

```
# 5.1.7. Figure 6: Chordal Graph for Streams
# Visualizing pairwise relationships between top two per stream, with the occupancy rate for the current period
# Intermittent stress & 0 & 0 & 0 & 0
# Intermittent stress & 0 & 0 & 0 & 0
# Intermittent stress & 0 & 0 & 0 & 0
# Intermittent stress & 0 & 0 & 0 & 0
# 0 & 0 & 0 & 0
# 0 & 0 & 0 & 0
#
```

```
# 2.5. Cluster and explore
parallelcoords(training2.z, gp.by.cluster=cluster$cluster, extra=TRUE)

group.size <- NULL
for (k in 1:k.max) {
  cluster <- kmeans(training2.z, k, nstart=20)
  group.size[k] <- cluster$size[1]
}
plot(1:k.max, group.size, type="l", ylim=c(0,nrow(training2.z)),
     xlab="Number of Clusters", ylab="Size of Smallest Cluster")
abline(h=0.05*nrow(training2.z), lty=2)

for (k in 2:k.max) {
  cluster <- kmeans(training2.z, k, nstart=20)
  parallelcoords(cluster$centers, extra=TRUE)
}

parallelcoords(training.cnts.mean.statistics$statistic)
# add the population mean to the plot (it should be at 0 in population-scaled)
abline(h=0,lwd=2)

parallelcoords(training.cnts.mean.statistics$statistic)
# add the population +/- 1.96*standard error to the plot
abline(h=1.96*training.cnts.mean.statistics$stderr, lwd=2, lty=2)
abline(h=-1.96*training.cnts.mean.statistics$stderr, lwd=2, lty=2)

parallelcoords(training.cnts.mean.statistics$statistic)
# add the population +/- 1.96*standard error to the plot
# (population-scaled)
abline(h=1.96/sqrt(nrow(training2.z)), lwd=2, lty=2)
abline(h=-1.96/sqrt(nrow(training2.z)), lwd=2, lty=2)

# 2.6. Compare cluster means to population means
cluster.means = apply(cluster$centers, 2, mean)
pop.means = apply(training2.z, 2, mean)
cbind(cluster.means, pop.means)

# 2.7. Compare cluster means to population means
# (population-scaled)
cluster.means.pop.scaled = apply(cluster$centers, 2, mean)
pop.means.pop.scaled = apply(training2.z, 2, mean)
cbind(cluster.means.pop.scaled, pop.means.pop.scaled)

# 2.8. Cluster membership variables
cluster.sizes = cbind(c(1:k.max), group.size)
colnames(cluster.sizes) = c("Number of Clusters", "Size of Smallest Cluster")
cluster.sizes

cluster.sizes[cluster.sizes[,2] > 0.05*nrow(training2.z),]
# select only those cluster sizes that are greater than 5%
cluster.sizes[cluster.sizes[,2] > 0.05*nrow(training2.z), 2, drop=FALSE]
# select only the second column, keep the data structure as a matrix
cluster.sizes[cluster.sizes[,2] > 0.05*nrow(training2.z), 2, drop=TRUE]
# select only the second column, drop the data structure to a vector

cluster.size.final = 3
cluster <- kmeans(training2.z, cluster.size.final, nstart=20)
table(cluster$cluster)

# 2.9. Compare cluster means
cluster.means = cbind(cluster$centers)
colnames(cluster.means) = c("Cluster 1", "Cluster 2", "Cluster 3")
cluster.means

cluster.means.t = t(cluster.means)
cluster.means.t[order(cluster.means.t[,1]),]
# sort by the first cluster

cluster.means.t[order(cluster.means.t[,2]),]
# sort by the second cluster

cluster.means.t[order(cluster.means.t[,3]),]
# sort by the third cluster

# 2.10. Compare cluster means to population means
cluster.means.pop = cbind(cluster.means.t, pop.means)
colnames(cluster.means.pop) = c("Cluster 1", "Cluster 2", "Cluster 3", "Population")
cluster.means.pop

cluster.means.pop.t = t(cluster.means.pop)
cluster.means.pop.t[order(cluster.means.pop.t[,1]),]
# sort by the first cluster

cluster.means.pop.t[order(cluster.means.pop.t[,2]),]
# sort by the second cluster

cluster.means.pop.t[order(cluster.means.pop.t[,3]),]
# sort by the third cluster

# 2.11. Cluster membership
cluster.plot = cbind(
  training$id,
  training$year,
  training$tract,
  training$cnty,
  training$tract.name,
  training$cnty.name,
  training$lat,
  training$lon,
  cluster$cluster)
colnames(cluster.plot) = c("id", "year", "tract", "cnty", "tract.name", "cnty.name", "lat", "lon", "cluster")
head(cluster.plot)

# 2.12. Export cluster membership
write.csv(cluster.plot, file="cluster.plot.csv")

# 2.13. Map cluster membership
# Install packages
install.packages("ggmap")
install.packages("ggplot2")
install.packages("mapproj")

# Load packages
library(ggmap)
library(ggplot2)
library(mapproj)

# Create a map of the study area
map <- qmap(c(lon=-122.3, lat=47.6), zoom=9, maptype="toner-lite")

# Map the cluster membership
map + geom_point(data=cluster.plot, aes(x=lon, y=lat, color=factor(cluster)))

# 2.14. Cluster profiles
# Create a data frame with the cluster membership and the training data
cluster.profiles <- data.frame(cluster.plot[,9], training)
colnames(cluster.profiles)[1] <- "cluster"
head(cluster.profiles)

# Create a table of the mean of each variable by cluster
round(aggregate(cluster.profiles, by=list(cluster.profiles$cluster), FUN=mean), digits=2)

# Create a table of the standard deviation of each variable by cluster
round(aggregate(cluster.profiles, by=list(cluster.profiles$cluster), FUN=sd), digits=2)

# Create a table of the number of observations in each cluster
table(cluster.profiles$cluster)

# 2.15. Cluster profile plots
# Install and load lattice
install.packages("lattice")
library(lattice)

# Create a boxplot for each variable by cluster
bwplot(cluster ~ id, data=cluster.profiles)
bwplot(cluster ~ year, data=cluster.profiles)
bwplot(cluster ~ tract, data=cluster.profiles)
bwplot(cluster ~ cnty, data=cluster.profiles)
bwplot(cluster ~ lat, data=cluster.profiles)
bwplot(cluster ~ lon, data=cluster.profiles)
bwplot(cluster ~ population, data=cluster.profiles)
bwplot(cluster ~ poverty, data=cluster.profiles)
bwplot(cluster ~ ownership, data=cluster.profiles)
bwplot(cluster ~ vacancy, data=cluster.profiles)
bwplot(cluster ~ mortgage, data=cluster.profiles)
bwplot(cluster ~ housing.age, data=cluster.profiles)
bwplot(cluster ~ median.rent, data=cluster.profiles)
bwplot(cluster ~ median.value, data=cluster.profiles)
bwplot(cluster ~ median.income, data=cluster.profiles)
bwplot(cluster ~ poverty.rate, data=cluster.profiles)
bwplot(cluster ~ ownership.rate, data=cluster.profiles)
bwplot(cluster ~ vacancy.rate, data=cluster.profiles)
bwplot(cluster ~ mortgage.rate, data=cluster.profiles)
bwplot(cluster ~ housing.age.rate, data=cluster.profiles)
bwplot(cluster ~ median.rent.rate, data=cluster.profiles)
bwplot(cluster ~ median.value.rate, data=cluster.profiles)
bwplot(cluster ~ median.income.rate, data=cluster.profiles)
bwplot(cluster ~ poverty.rate.rate, data=cluster.profiles)
bwplot(cluster ~ ownership.rate.rate, data=cluster.profiles)
bwplot(cluster ~ vacancy.rate.rate, data=cluster.profiles)
bwplot(cluster ~ mortgage.rate.rate, data=cluster.profiles)
bwplot(cluster ~ housing.age.rate.rate, data=cluster.profiles)
bwplot(cluster ~ median.rent.rate.rate, data=cluster.profiles)
bwplot(cluster ~ median.value.rate.rate, data=cluster.profiles)
bwplot(cluster ~ median.income.rate.rate, data=cluster.profiles)

# 2.16. Cluster profile tables
# Create a table of the mean of each variable by cluster
round(aggregate(cluster.profiles, by=list(cluster.profiles$cluster), FUN=mean), digits=2)

# Create a table of the standard deviation of each variable by cluster
round(aggregate(cluster.profiles, by=list(cluster.profiles$cluster), FUN=sd), digits=2)

# Create a table of the number of observations in each cluster
table(cluster.profiles$cluster)

# 2.17. Cluster profile maps
# Create a data frame with the cluster membership and the training data
cluster.profiles <- data.frame(cluster.plot[,9], training)
colnames(cluster.profiles)[1] <- "cluster"
head(cluster.profiles)

# Create a map of the study area
map <- qmap(c(lon=-122.3, lat=47.6), zoom=9, maptype="toner-lite")

# Map the cluster membership
map + geom_point(data=cluster.profiles, aes(x=lon, y=lat, color=factor(cluster)))

# 2.18. Export cluster profiles
write.csv(cluster.profiles, file="cluster.profiles.csv")

# 2.19. Cluster profile tables
# Create a table of the mean of each variable by cluster
round(aggregate(cluster.profiles, by=list(cluster.profiles$cluster), FUN=mean), digits=2)

# Create a table of the standard deviation of each variable by cluster
round(aggregate(cluster.profiles, by=list(cluster.profiles$cluster), FUN=sd), digits=2)

# Create a table of the number of observations in each cluster
table(cluster.profiles$cluster)

# 2.20. Cluster profile plots
# Install and load lattice
install.packages("lattice")
library(lattice)

# Create a boxplot for each variable by cluster
bwplot(cluster ~ id, data=cluster.profiles)
bwplot(cluster ~ year, data=cluster.profiles)
bwplot(cluster ~ tract, data=cluster.profiles)
bwplot(cluster ~ cnty, data=cluster.profiles)
bwplot(cluster ~ lat, data=cluster.profiles)
bwplot(cluster ~ lon, data=cluster.profiles)
bwplot(cluster ~ population, data=cluster.profiles)
bwplot(cluster ~ poverty, data=cluster.profiles)
bwplot(cluster ~ ownership, data=cluster.profiles)
bwplot(cluster ~ vacancy, data=cluster.profiles)
bwplot(cluster ~ mortgage, data=cluster.profiles)
bwplot(cluster ~ housing.age, data=cluster.profiles)
bwplot(cluster ~ median.rent, data=cluster.profiles)
bwplot(cluster ~ median.value, data=cluster.profiles)
bwplot(cluster ~ median.income, data=cluster.profiles)
bwplot(cluster ~ poverty.rate, data=cluster.profiles)
bwplot(cluster ~ ownership.rate, data=cluster.profiles)
bwplot(cluster ~ vacancy.rate, data=cluster.profiles)
bwplot(cluster ~ mortgage.rate, data=cluster.profiles)
bwplot(cluster ~ housing.age.rate, data=cluster.profiles)
bwplot(cluster ~ median.rent.rate, data=cluster.profiles)
bwplot(cluster ~ median.value.rate, data=cluster.profiles)
bwplot(cluster ~ median.income.rate, data=cluster.profiles)
bwplot(cluster ~ poverty.rate.rate, data=cluster.profiles)
bwplot(cluster ~ ownership.rate.rate, data=cluster.profiles)
bwplot(cluster ~ vacancy.rate.rate, data=cluster.profiles)
bwplot(cluster ~ mortgage.rate.rate, data=cluster.profiles)
bwplot(cluster ~ housing.age.rate.rate, data=cluster.profiles)
bwplot(cluster ~ median.rent.rate.rate, data=cluster.profiles)
bwplot(cluster ~ median.value.rate.rate, data=cluster.profiles)
bwplot(cluster ~ median.income.rate.rate, data=cluster.profiles)

# 2.21. Export training data
write.csv(training, file="training.csv")

# 2.22. Export training2 data
write.csv(training2, file="training2.csv")

# 2.23. Export training2.z data
write.csv(training2.z, file="training2.z.csv")

# 2.24. Export training.cnts.mean.statistics data
write.csv(training.cnts.mean.statistics$statistic, file="training.cnts.mean.statistics.csv")
write.csv(training.cnts.mean.statistics$stderr, file="training.cnts.mean.statistics.stderr.csv")

# 2.25. Export cluster data
write.csv(cluster, file="cluster.csv")

# 2.26. Export cluster.plot data
write.csv(cluster.plot, file="cluster.plot.csv")

# 2.27. Export cluster.profiles data
write.csv(cluster.profiles, file="cluster.profiles.csv")
```

3.0 Exploratory Analysis

3.1 Cluster Profiles

Clusters are filled where cluster more is higher price, higher commercial mix areas, while **green** clusters are more common where commercial mix is lower. In contrast, the three lower overlap much more in building coverage.

Overall, parking area and commercial mix distinguish parking patterns better than building coverage.

```
# 3.1. Cluster distributions of exploratory variables
# Create a boxplot for each variable by cluster
bwplot(cluster~median.value|factor("Cluster Labels"), ylab="Cluster",
       xlab="Median Value",
       main="2010")

pdf(file="pdf.name.pdf", 7, 7)
  "all_lat_lon_map.pdf",
  "commercial_mixed_index.pdf",
  "building_coverage.pdf",
  "parking_space.pdf",
  "architectural_firmspan.pdf",
  "architectural_stories.pdf",
  "population.count.pdf",
  "architectural_lot.frontspan.pdf",
  "speed.limit.pdf",
  "architectural_stories.thousandthousand.pdf",
  "street.width.pdf",
  "transport.parking.space.pdf",
  "area.vacant.pdf",
  "residential_shared.pdf"
)

# 3.2. Figure 10 PDPs
# Create a data frame with the variables
variables.features <-
  "all_lat_lon_map",
  "commercial_mixed_index",
  "building_coverage",
  "parking_space",
  "architectural_firmspan",
  "architectural_stories",
  "population.count",
  "architectural_lot.frontspan",
  "speed.limit",
  "architectural_stories.thousandthousand",
  "street.width",
  "transport.parking.space",
  "area.vacant",
  "residential_shared"
)

features.pdf.1 <- NA # to contain features of 1 in pdf.df column

features.pdf.2 <-
  c.fmr # to find values, "architectural" gt 1 in pdf.df column

pdf.df <- cbind(features.pdf.1, features.pdf.2, categorical.pdf.1, class)
colnames(pdf.df) <-
  "features.pdf.1",
  "features.pdf.2",
  "categorical.pdf.1",
  "class"
)

pdf.df <- pdf.df[order(pdf.df$class, -pdf.df$features.pdf.2),]
pdf.df
# sort by the class, then by the second feature, descending

pdf.df[,1] <- as.character(pdf.df[,1])
pdf.df[,2] <- as.numeric(pdf.df[,2])
pdf.df[,3] <- as.numeric(pdf.df[,3])
pdf.df[,4] <- as.factor(pdf.df[,4])
pdf.df[,5] <- as.factor(pdf.df[,5])
pdf.df
# check the data types

pdf.df[pdf.df$class=="Commercial", ]
# select only the commercial class

pdf.df[pdf.df$class=="Commercial", 1:2]
# select only the first two features

pdf.df[pdf.df$class=="Commercial", 1:2, drop=FALSE]
# select only the first two features, keep the data structure as a matrix

pdf.df[pdf.df$class=="Commercial", 1:2, drop=TRUE]
# select only the first two features, drop the data structure to a vector
```